

Dummit & Foote: Exercises

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0 Preliminaries

0.1 Basics

For Exercises 1 to 4, let

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and $\mathcal{B} = \{X \in \mathcal{A} : MX = XM\}$, where $\mathcal{A}$ denotes the set of 2×2 matrices with real entries.

(1) Determine which of the following elements of $\mathcal{A}$ lie in $\mathcal{B}$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. Note that the 1st matrix is M itself so that it belongs to $\mathcal{B}$, since $MX = XM = M^2$. Further note that the 3rd, 5th, and 6th matrices are the zero, identity, and exchange matrices respectively. Then the 3rd and 5th belong to $\mathcal{B}$ while the 6th does not. Then only the remaining matrices to check are the 2nd and 4th matrices. For the 2nd matrix:

$$\begin{aligned} MX &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 1 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \\ XM &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \end{aligned}$$

Then $MX \neq XM$ and the 2nd matrix does not belong to $\mathcal{B}$. For the 4th matrix:

$$\begin{aligned} MX &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 1 + 1 \cdot 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \\ XM &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

So that $MX \neq XM$. □

(2) Prove that if $P, Q \in \mathcal{B}$, then $P + Q \in \mathcal{B}$ where $+$ denotes the usual sum of two matrices.

Solution. For Exercises 2 and 3, let

$$P = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad Q = \begin{pmatrix} e & f \\ g & h \end{pmatrix}, \quad P + Q = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}, \quad P \cdot Q = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

where $PM = MP$ and $QM = MQ$. Then

$$\begin{aligned} M(P+Q) &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix} \\ &= \begin{pmatrix} a+e+c+g & b+f+d+h \\ c+g & d+h \end{pmatrix} \\ &= \begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix} + \begin{pmatrix} e+g & f+h \\ g & h \end{pmatrix} \\ &= MP + MQ \\ &= PM + QM \\ &= \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix} + \begin{pmatrix} e & e+f \\ g & g+h \end{pmatrix} \\ &= \begin{pmatrix} a+e & a+e+b+f \\ c+g & c+g+d+h \end{pmatrix} \\ &= \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= (P+Q)M \end{aligned}$$

□

- (3) Prove that if $P, Q \in \mathcal{B}$, then $P \cdot Q \in \mathcal{B}$ where $\cdot$ denotes the usual product of two matrices.

Solution. Using PQ , proceed as we did above with $M(PQ)$. Rewriting the entries will result in $(MP)Q$. Since $P \in \mathcal{B}$, then we have $(PM)Q$. We rewrite entries again to result in $P(MQ)$. Because $Q \in \mathcal{B}$, we have $P(QM)$, and a final rewrite results in $(PQ)M$. $\square$

- (4) Find conditions on p, q, r, s which determine precisely when $\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathcal{B}$.

Solution. Let X be the matrix described above. Note that

$$\begin{aligned} MX &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p+r & q+s \\ r & s \end{pmatrix} \\ XM &= \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & p+q \\ r & r+s \end{pmatrix} \end{aligned}$$

Because $X \in \mathcal{B}$, then we may compare entries to obtain the following:

$$\begin{cases} p+r=p \\ q+s=p+q \\ r=r \\ s=r+s \end{cases}$$

The first and fourth equations force $r=0$, and the second equation forces $p=s$. Then $\mathcal{B}$ is classified as

$$\mathcal{B} = \left\{ \begin{pmatrix} p & p+q \\ 0 & p \end{pmatrix} \mid p, q \in \mathbb{R} \right\} \quad \square$$

- (5) Determine whether the following functions f are well defined:

- (a) $f: \mathbb{Q} \rightarrow \mathbb{Z}$ defined by $f(a/b) = a$.
 (b) $f: \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(a/b) = a^2/b^2$.

Solution.

- (a) No, because $1/2 = 2/4$, but $f(1/2) = 1$ and $f(2/4) = 2$.
 (b) Yes; suppose $a/b = c/d$. Then $ad = bc$, or that $a^2d^2 = b^2c^2$. Then $a^2/b^2 = c^2/d^2$, or $f(a/b) = f(c/d)$. $\square$

- (6) Determine whether the function $f: \mathbb{R}^+ \rightarrow \mathbb{Z}$ defined by mapping a real number r to the first digit to the right of the decimal point in a decimal expansion of r is well defined.

Solution. No. Note that $1 = 1.000\dots = 0.999\dots$, but $f(1.000\dots) = 1$ and $f(0.999\dots) = 9$. $\square$

- (7) Let $f: A \rightarrow B$ be a surjective map of sets. Prove that the relation

$$a \sim b \iff f(a) = f(b)$$

is an equivalence relation whose equivalence classes are the fibers of f .

Solution. The relation above is an equivalence relation, since $=$ is already an equivalence relation on B .

Consider now some $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. The equivalence class of a is the set $\{x \in A \mid x \sim a\}$. By definition of $\sim$, this is equal to the set $\{x \in A \mid f(x) = f(a) = b\}$, which is precisely the fiber of f over b . $\square$

0.2 Properties of the Integers

1. For each of the following pairs of integers a and b , determine their greatest common divisor, their least common multiple, and write their greatest common divisor in the form $ax + by$ for some integers x and y .

- (a) $a = 20, b = 13$
- (b) $a = 69, b = 372$
- (c) $a = 792, b = 275$
- (d) $a = 11391, b = 5673$
- (e) $a = 1761, b = 1567$
- (f) $a = 507885, b = 60808$

Solution.

- (a) Note: for this exercise, we will only do (e) as that has the most steps in calculating both the g.c.d and the Euclidean Algorithm. The l.c.m is obtained by dividing the product ab by (a, b) .

$$(20, 13) = 1, \text{lcm}(20, 13) = 260, 1 = 2(20) - 3(13)$$

$$(b) (69, 372) = 3, \text{lcm}(69, 372) = 8556, 27(69) - 5(372)$$

$$(c) (792, 275) = 11, \text{lcm}(792, 275) = 19800, 8(792) - 23(275)$$

$$(d) (11391, 5673) = 3, \text{lcm}(11391, 5673) = 21540381, 3 = 253(5673) - 126(11391)$$

- (e) Applying the Euclidean Algorithm to $a = 1761$ and $b = 1567$, we get:

$$1761 = (1)1567 + 194$$

$$1567 = (8)194 + 15$$

$$194 = (12)15 + 14$$

$$15 = (1)14 + 1$$

Then $(1761, 1567) = 1$ so that $\text{lcm}(1761, 1567) = 2759487$. Reversing the Euclidean Algorithm steps to solve for 1, we get:

$$\begin{aligned} 1 &= 15 - 14 \\ &= 15 - (194 - 12(15)) = 13(15) - 194 \\ &= 13(1567 - 8(194)) - 194 = 13(1567) - 105(194) \\ &= 13(1567) - 105(1761 - 1567) = -105(1761) + 118(1567) \end{aligned}$$

$$(f) (507885, 60808) = 691, \text{lcm}(507885, 60808) = 44693880, 691 = 142(60808) - 17(507885) \quad \square$$

2. Prove that if the integer k divides the integers a and b then k divides $as + bt$ for every pair of integers s and t .

Solution. Since $k \mid a$ and $k \mid b$, there exists $x, y \in \mathbb{Z}$ such that $a = kx$ and $b = ky$. Then for any $s, t \in \mathbb{Z}$, we have $as + bt = kxs + kyt = k(xs + yt)$ which is divisible by k . $\square$

3. Prove that if n is composite then there are integers a and b such that n divides ab but n does not divide either a or b .

Solution. By the Fundamental Theorem of Arithmetic, n has at least two prime factors a, b such that $1 < a, b < n$. Putting $n = ab$, then $n \mid ab$ but $n \nmid a$ and $n \nmid b$. $\square$

4. Let a, b and N be fixed integers with a and b nonzero and let $d = (a, b)$ be the greatest common divisor of a and b . Suppose x_0 and y_0 are particular solutions to $ax + by = N$ (i.e., $ax_0 + by_0 = N$). Prove for any integer t that the integers

$$x = x_0 + \frac{b}{d}t \text{ and } y = y_0 - \frac{a}{d}t$$

are also solutions to $ax + by = N$ (this is in fact the general solution).

Solution. We have

$$\begin{aligned} ax + by &= a\left(x_0 + \frac{b}{d}t\right) + b\left(y_0 - \frac{a}{d}t\right) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = N \end{aligned}$$

$\square$

5. Determine the value $\varphi(n)$ each integer $n \leq 30$ where φ denotes the Euler φ -function.

Solution.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	8	16	6	18	8	12	10	22	8	20	12	18	12	28	8

□

6. Prove the Well Ordering Property of $\mathbb{Z}$ by induction and prove the minimal element is unique.

Solution. Let $A \subseteq \mathbb{N}$ be nonempty. If $0 \in A$, then it has a minimal element. Suppose now that $0 \notin A$. Moreover, suppose that $1, 2, \dots, k \notin A$ for some k . By strong induction, it follows that $k+1 \notin A$. However, this results in there being no positive integer in A , contradicting that it is nonempty, thus it must have a minimal element. Moreover, if it had two minimal elements a, b , then it follows that $a \leq b$ and $b \leq a$ by definition of minimal element. Hence, $a = b$, and the minimal element of A is unique. □

7. If p is a prime, prove that there do not exist nonzero integers a and b such that $a^2 = pb^2$ (i.e., $\sqrt{p}$ is not a rational number).

Solution. Suppose $\sqrt{p}$ is a rational number. Then there exists $a, b \in \mathbb{Z}$ such that $a^2 = b^2 p$, and $(a, b) = 1$. If $(a, b) = d \neq 1$, then take $a' = a/d$ and $b' = b/d$ instead. Then $p \mid a^2 = a \cdot a$, and $p \mid a$. We then have $a = kp$ for some $k \in \mathbb{Z}$. Then $b^2 = k^2 p$ so that $p \mid b$, and $b = mp$ for $m \in \mathbb{Z}$. This contradicts that $(a, b) = 1$, hence $\sqrt{p}$ cannot be a rational number. □

8. Let p be a prime, $n \in \mathbb{N}$. Find a formula for the largest power of p which divides $n! = n(n-1)(n-2)\dots 2 \cdot 1$ (it involves the greatest integer function).

Solution. Note that in $n!$, there are $\lfloor n/p \rfloor$ integers that are divisible by p , where the greatest integer function is necessary since n may not be a perfect multiple of p (consider $n = 36$ and $p = 7$. Then there are the multiples 7, 14, 21, 28, and 35 which is $\lfloor 36/7 \rfloor = 5$). However, this alone does not account for the factors of p in $n!$, so we move onto p^2 . In this case, there are $\lfloor n/p^2 \rfloor$ integers divisible by p^2 . We continue this process until a certain integer $a \in \mathbb{N}$ such that $p^a \leq n$. (We must bound p^a above by n , up to equality, since n may be a power of p , and any power a such that $p^a > n$ would result in $\lfloor n/p^a \rfloor = 0$). It then follows that $a \leq \log_p(n)$, or that the maximum power of p (and any of its multiples) that divide $n!$ is given by $a = \lfloor \log_p(n) \rfloor$. Thus, the largest power of p that divides $n!$ is given by

$$\sum_{i=1}^{\lfloor \log_p(n) \rfloor} \left\lfloor \frac{n}{p^i} \right\rfloor$$

□

9. Write a computer program to determine the greatest common divisor (a, b) of two integers a and b and to express (a, b) in the form $ax + by$ for some integers x and y .

Require: Integers a, b (not both zero)

Ensure: Integers g, x, y such that $g = (a, b)$ and $ax + by = g$

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1:  $x_1 \leftarrow 1, y_1 \leftarrow 0$ 
2:  $x_2 \leftarrow 0, y_2 \leftarrow 1$ 
3: while  $b \neq 0$  do
4:    $q \leftarrow 0$ 
5:    $r \leftarrow a$ 
6:   while  $(r \geq b)$  or  $(r \leq -b)$  do
7:      $r \leftarrow r - b$ 
8:      $q \leftarrow q + 1$ 
9:   end while
10:   $a \leftarrow b$ 
11:   $b \leftarrow r$ 

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12:    $(x_1, x_2) \leftarrow (x_2, x_1 - q \times x_2)$ 
13:    $(y_1, y_2) \leftarrow (y_2, y_1 - q \times y_2)$ 
14: end while
15:  $g \leftarrow a$ 
16: return  $(g, x_1, y_1)$ 

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10. Prove for any given positive integer N there exist only finitely many integers n with $\varphi(n) = N$ where φ denotes Euler's φ -function. Conclude in particular that φ tends to infinity as n tends to infinity.

Solution. Let $N \in \mathbb{N}$ be fixed, and let $n \in \mathbb{N}$ such that $\varphi(n) = N$. We can break apart n into its primes p_i (where $p_1 < p_2 < \dots$) with associated exponents α_i :

$$n = \prod_{i=1}^k p_i^{\alpha_i} \implies \varphi(n) = \prod_{i=1}^k p_i^{\alpha_i-1} (p_i - 1) = N$$

In particular, each of the terms $p_i - 1$ divides N so that $p_i - 1 \leq N$, or $p_i \leq N + 1$. It follows that any prime q of some n must be such a prime where $q - 1 \leq N$. Moreover, for some $p_i^{\alpha_i}$ that is a part of n 's prime factorization, it follows that $p_i^{\alpha_i-1}$ divides N . Since $\alpha_i - 1$ is finite, this limits the exponents for any particular p_i . With a finite list of primes and finitely many exponents, it follows that the amount of n such that $\varphi(n) = N$ is finite.

For some $N_0 \in \mathbb{N}$, there are finitely many n such that $\varphi(n) = N_0$. Then there are infinitely many $m > n$ such that $\varphi(m) > N_0$, or that $\varphi(n)$ tends towards infinity. $\square$

11. Prove that if d divides n then $\varphi(d)$ divides $\varphi(n)$ where φ denotes Euler's φ -function.

Solution. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ with $d \mid n$. Then d is a composition of some p_i present in n , so some α_i may go to 0 or are less. In particular, we have that $d = p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}$, where $\beta_i \leq \alpha_i$ for all i . To see if $\varphi(d) \mid \varphi(n)$, it follows to check if $p_i^{\beta_i-1} (p_i - 1)$ divides $p_i^{\alpha_i-1} (p_i - 1)$, which further simplifies to checking if $p_i^{\beta_i}$ divides $p_i^{\alpha_i}$ for each i . Clearly, $p_i^{\alpha_i} = p_i^{\beta_i} \cdot p_i^{\alpha_i-\beta_i}$, hence $\varphi(d) \mid \varphi(n)$. $\square$